

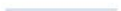
Feature Engineering by Data Type

Encode domain knowledge without leaking information across model boundaries.



Learning outcomes

01  Transform numeric, categorical, temporal, and text fields.

02  Recognize target leakage.

03  Compare engineered features against a baseline.

The concept system

01

encoding

02

scaling

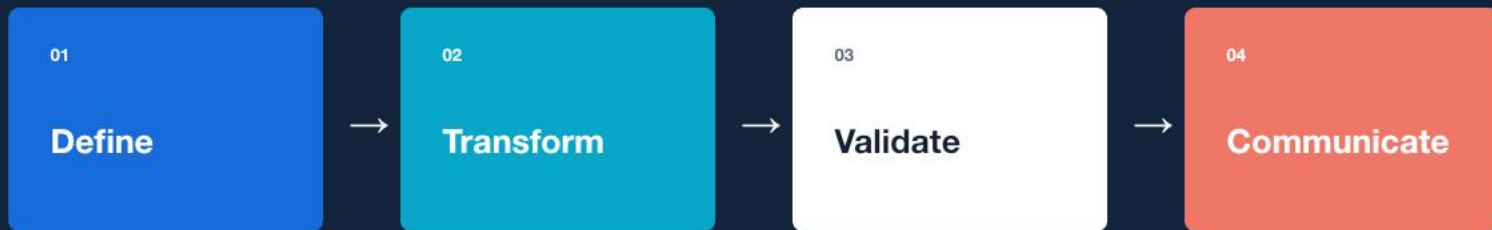
03

time features

04

leakage

A repeatable reasoning loop



Engineer behavioral features for a churn prediction problem.

Guided lab

LAB BRIEF

**Engineer behavioral features
for a churn prediction
problem.**

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

A feature dictionary with rationale and leakage review.

QUALITY GATE

-  **Reproducible**
-  **Tested**
-  **Decision-relevant**

WEEKLY ASSESSMENT

Leakage-Safe Feature Pipeline

Boundary correctness 35%,
rationale 25%, reproducibility
20%, tests 10%, explanation
10%.

What should remain after the lesson?

CONCEPT

encoding + scaling

PRACTICE

Engineer behavioral features for a churn prediction problem.

EVIDENCE

A feature dictionary with rationale and leakage review.
